

Computer Networks

Network Security

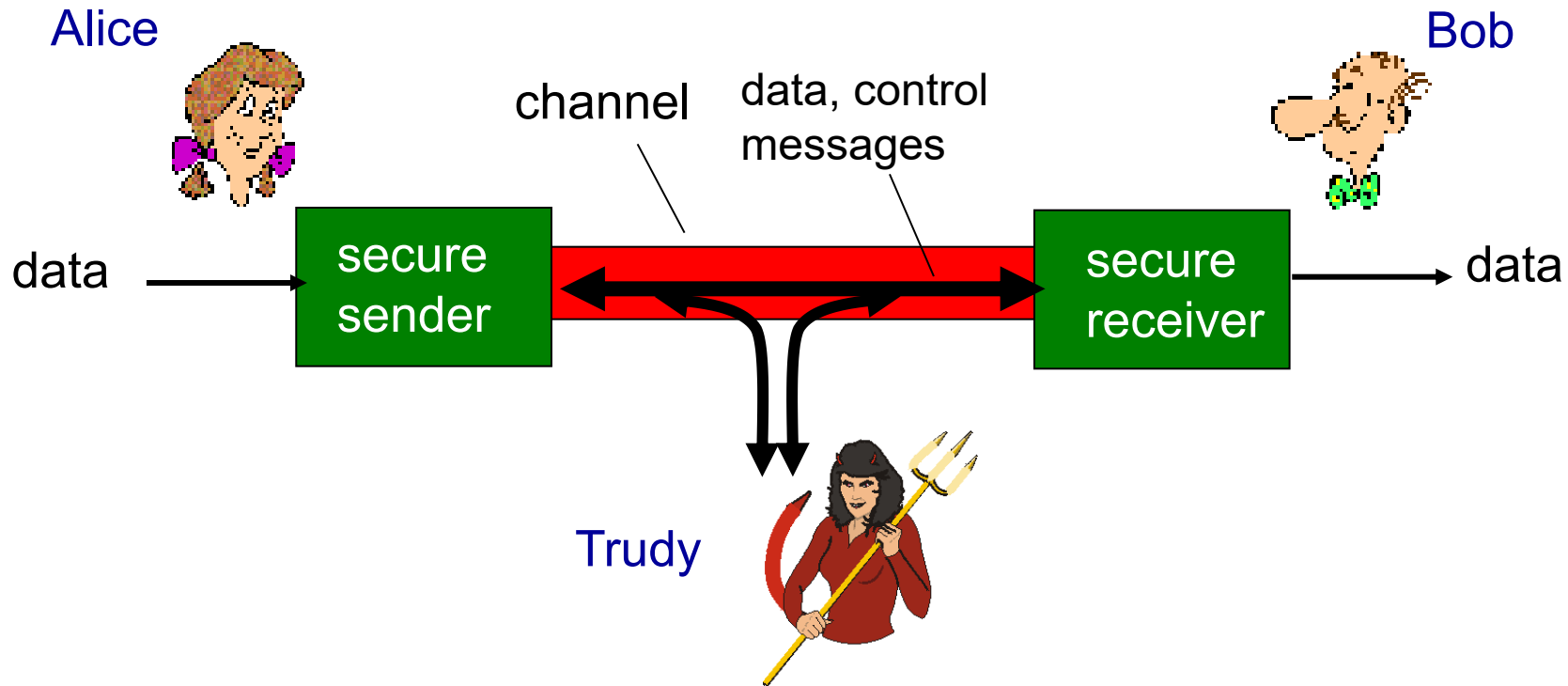
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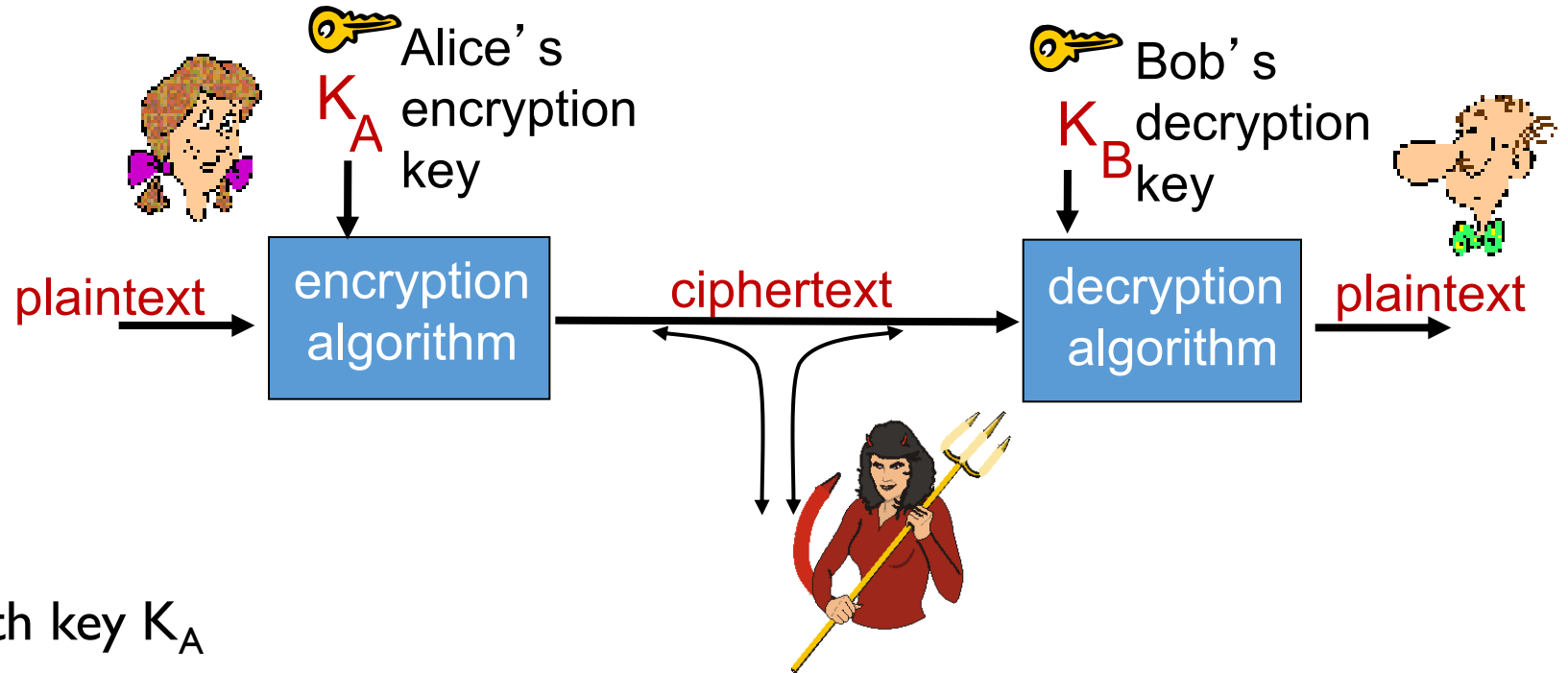
Friends and enemies: Alice, Bob, Trudy

- Bob, Alice want to communicate “securely”
- Trudy (intruder) may intercept, delete, add messages



Principles of Cryptography

The language of cryptography

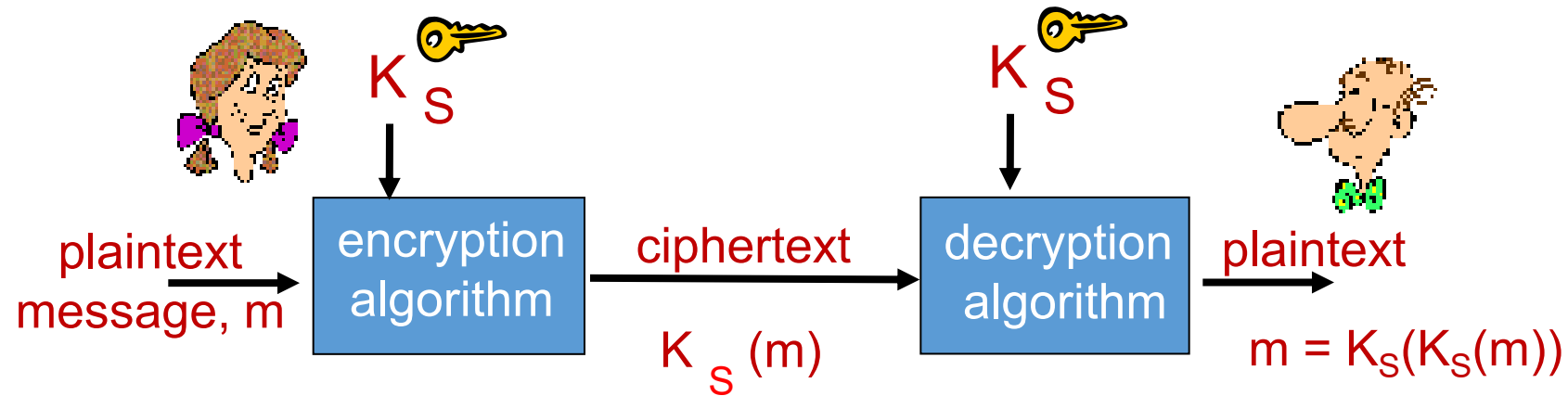


m plaintext message

$K_A(m)$ ciphertext, encrypted with key K_A

$m = K_B(K_A(m))$

Symmetric key cryptography



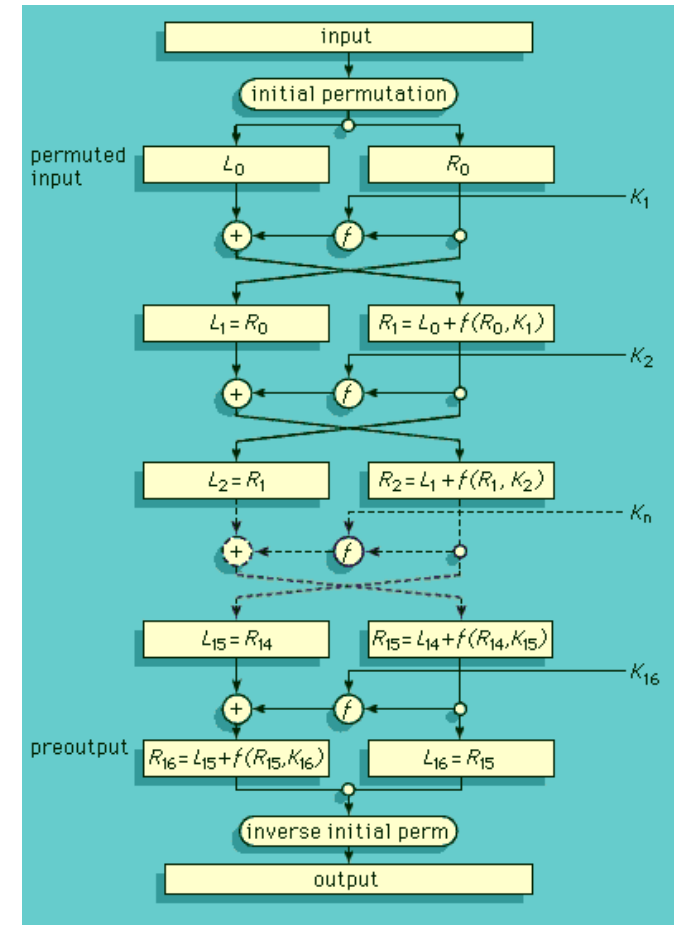
symmetric key crypto: Bob and Alice share same (symmetric) key: K_S

Examples: DES (Data Encryption Standard), AES (Advanced Encryption Standard)

Symmetric key crypto: DES

DES: Data Encryption Standard

- US encryption standard [NIST 1993]
- 56-bit symmetric key, 64-bit plaintext input
- block cipher with cipher block chaining
- making DES more secure:
 - 3DES: encrypt 3 times with 3 different keys



Src: <https://www.britannica.com/topic/Data-Encryption-Standard>

AES: Advanced Encryption Standard

- symmetric-key NIST standard, replaced DES (Nov 2001)
 - processes data in 128 bit blocks
 - 128, 192, or 256 bit keys
 - brute force decryption (try each key) taking 1 sec on DES, takes 149 trillion years for AES
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Public Key Cryptography

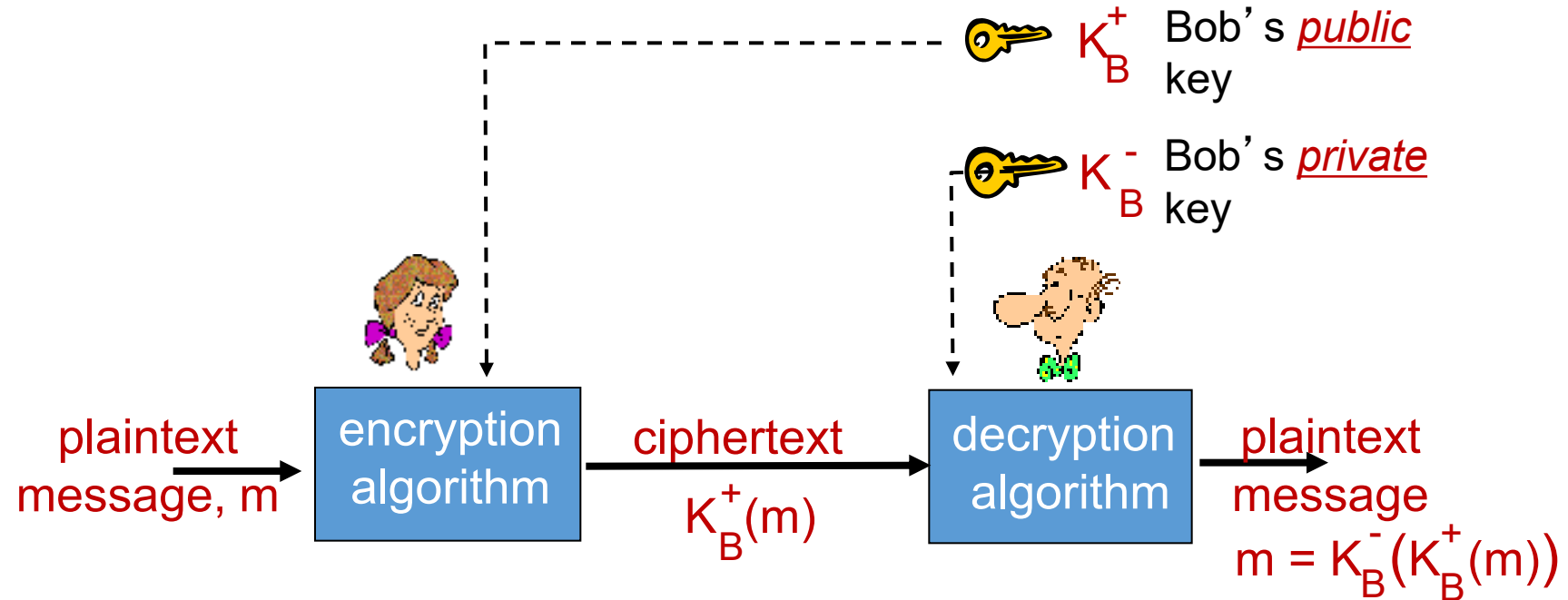
symmetric key crypto

- requires sender, receiver know shared secret key
- Q: how to agree on key in first place (particularly if never “met”)?

public key crypto

- radically different approach [Diffie-Hellman76, RSA78]
- sender, receiver do *not* share secret key
- *public* encryption key known to *all*
- *private* decryption key known only to receiver

Public key cryptography



Public key encryption algorithms

① need $K_B^+(\cdot)$ and $K_B^-(\cdot)$ such that

$$K_B^-(K_B^+(m)) = m$$

② given public key K_B^+ , it should be impossible to compute private key K_B^-

RSA: Rivest, Shamir, Adelson algorithm

RSA: Creating public/private key pair

1. choose two large prime numbers p, q . (e.g., 1024 bits each)
2. compute $n = pq$, $z = (p-1)(q-1)$
3. choose e (with $e < n$) that has no common factors with z (e, z are “relatively prime”).
4. choose d such that $ed-1$ is exactly divisible by z . (in other words: $ed \bmod z = 1$).
5. public key is (n, e) . private key is (n, d) .

K_B^+

K_B^-

RSA: encryption, decryption

0. given (n,e) and (n,d) as computed above

1. to encrypt message m ($<n$), compute

$$c = m^e \bmod n$$

2. to decrypt received bit pattern, c , compute

$$m = c^d \bmod n$$

magic happens!

$$m = \underbrace{(m^e \bmod n)}_c^d \bmod n$$

$$\text{Why? } m = \underbrace{(m^e \bmod n)}_c^d \bmod n$$

- Useful number theory result: If p, q are prime and $n = pq$, then $x^y \bmod n = x^{y \bmod (p-1)(q-1)} \bmod n$
 - $(m^e \bmod n)^d \bmod n$
 - $= m^{ed} \bmod n$
 - $= m^{ed \bmod (p-1)(q-1)} \bmod n$ [using the theorem]
 - $= m^1 \bmod n$ [as $ed-1$ is divisible by $(p-1)(q-1)$]
 - $= m$
-

RSA: another important property

The following property will be *very* useful later:

$$\underbrace{K_B^-(K_B^+(m))}_{\text{use public key first, followed by private key}} = m = \underbrace{K_B^+(K_B^-(m))}_{\text{use private key first, followed by public key}}$$

use public key first,
followed by
private key

use private key
first, followed by
public key

Why $K_B^-(K_B^+(m)) = m = K_B^+(K_B^-(m))$?

follows directly from modular arithmetic:

$$\begin{aligned}(m^e \bmod n)^d \bmod n &= m^{ed} \bmod n \\ &= m^{de} \bmod n \\ &= (m^d \bmod n)^e \bmod n\end{aligned}$$

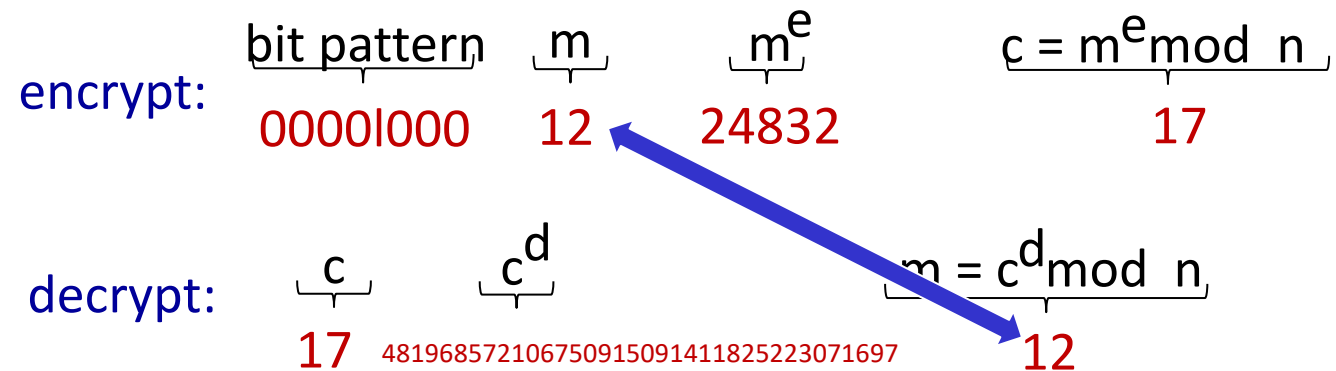
RSA example:

Bob chooses $p=5$, $q=7$. Then $n=35$, $z=24$.

$e=5$ (so e, z relatively prime).

$d=29$ (so $ed-1$ exactly divisible by z).

encrypting 8-bit messages.



RSA in practice: session keys

- exponentiation in RSA is computationally intensive
- DES is at least 100 times faster than RSA
- use public key crypto to establish secure connection, then establish second key – symmetric session key – for encrypting data

session key, K_s

- Bob and Alice use RSA to exchange a symmetric session key K_s
 - once both have K_s , they use symmetric key cryptography
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